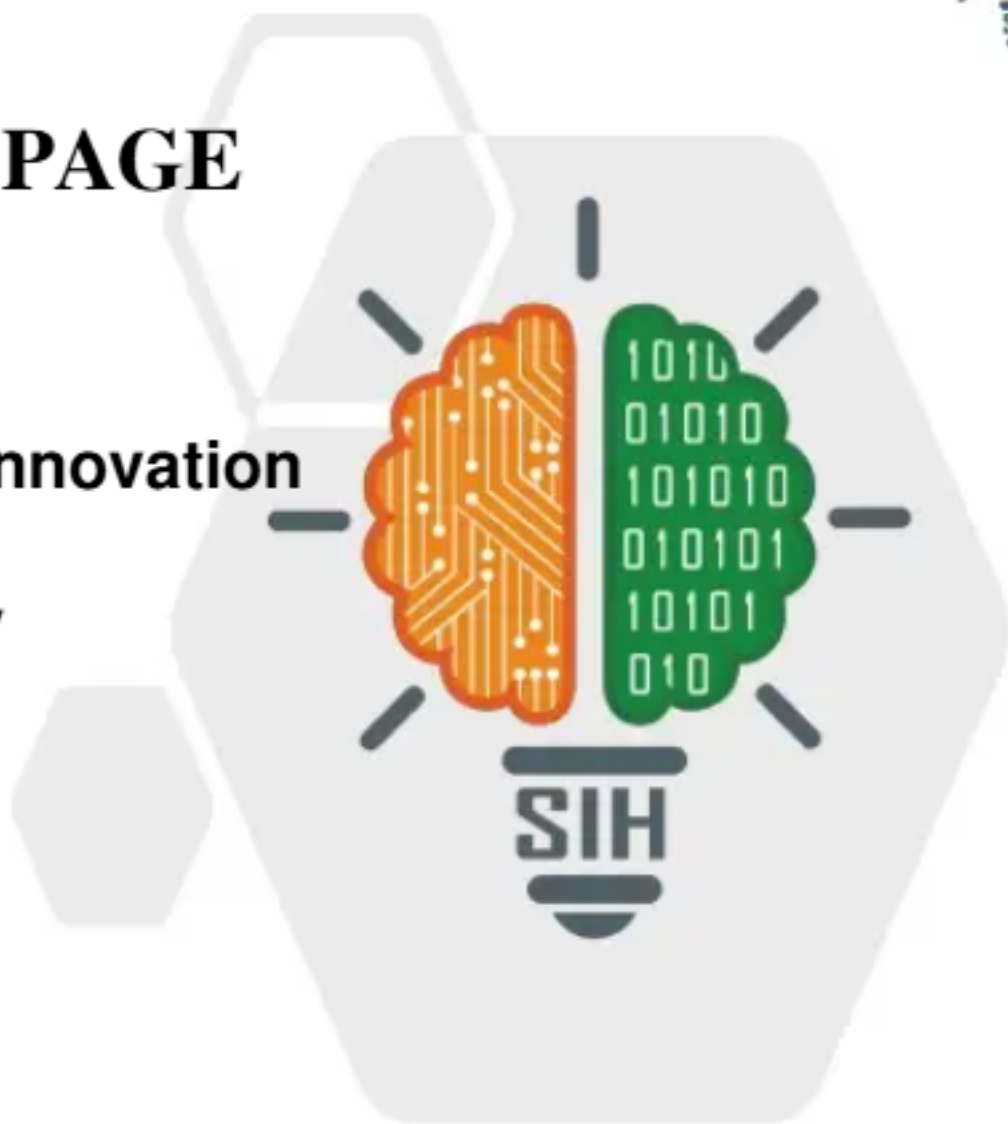


SMART INDIA HACKATHON 2024



TITLE PAGE

- Problem Statement ID - SIH1592
- Problem Statement Title- Student Innovation
- Theme- Clean & Green Technology
- PS Category- Software
- Team ID-
- Team Name- CleanCoders



Idea

- The Problem

According to literature reports nearly 65% of waste can be collected by municipality, but there is a lack of a proper real-time waste-collecting mechanism, due to which only 35% of waste can not be collected by municipality

- The Need

To maintain a clean and green environment we need a smart waste management and classification system.

- The Proposed Solution

Our proposal involves real-time smart waste management using user-submitted images, tracked via Google Maps. Municipality workers promptly collect garbage, and efficient workers receive rewards.

2

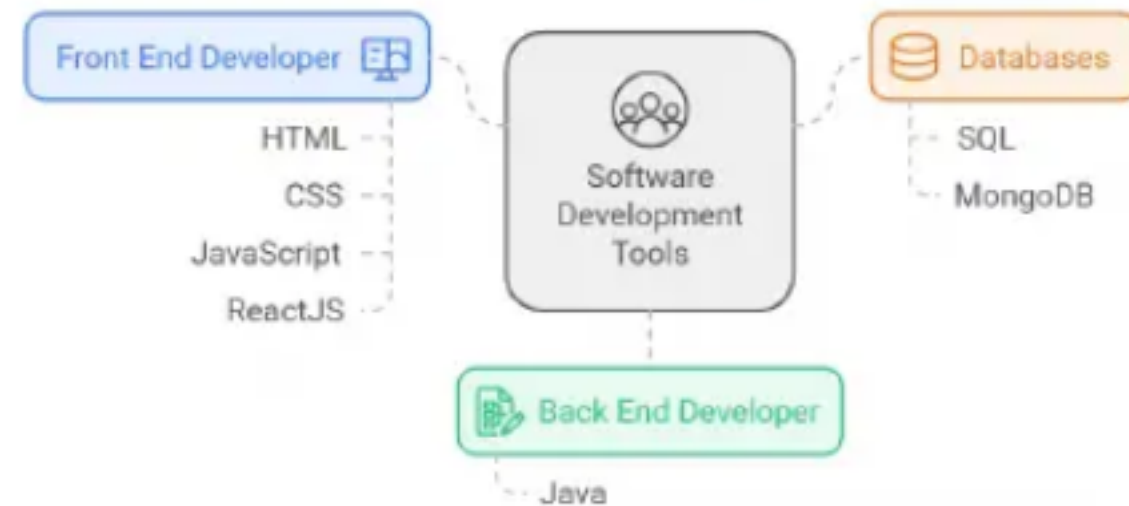


TECHNICAL APPROACH



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Technology stack:

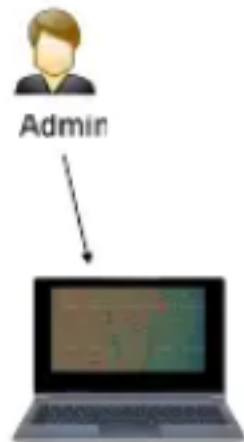


Waste sorting:-

Key Components in IoT for Waste Sorting



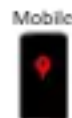
Clean
Coders



Collect the Waste



Mark the Waste location



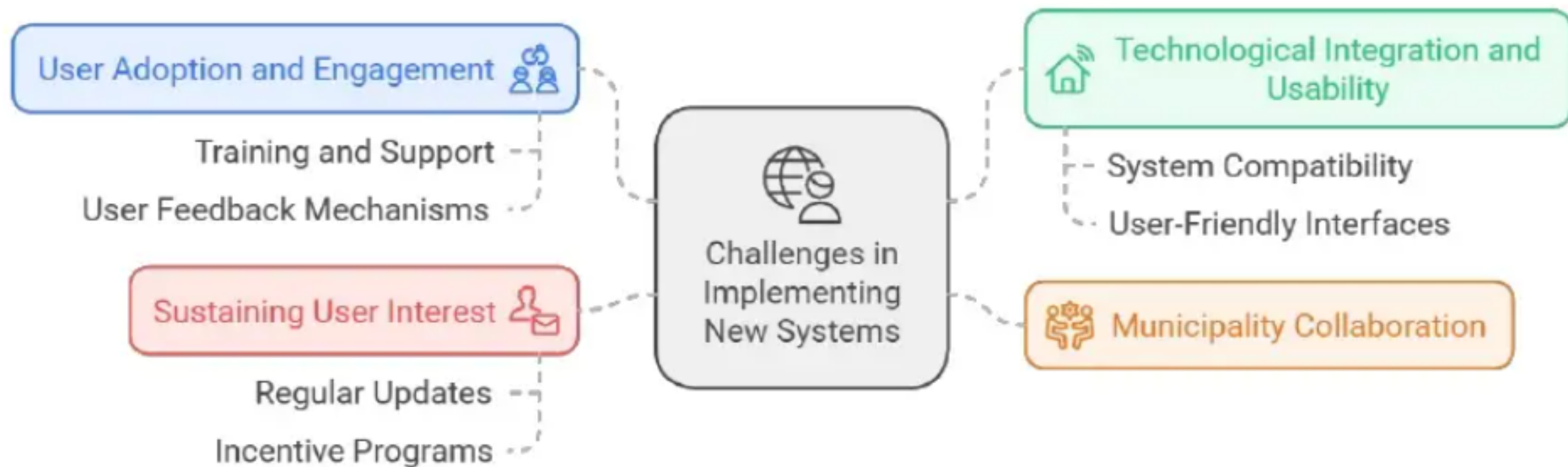
Mobile



Waste Vehicle

- We propose a real-time smart waste management system where users submit images of local garbage through a dedicated platform. The system classifies the waste and notifies municipal workers for collection within two days. Additionally, users can earn money by selling recyclable materials, promoting recycling and improving waste management efficiency.

Strategies for overcoming these challenges



IMPACT AND BENEFITS



RESEARCH AND REFERENCES



- <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9566108/>
- [H16.pdf \(iitk.ac.in\)](#)
- www.excellentwebworld.com/smart-waste-management-solution